Shlok Channawar

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Education

The Pennsylvania State University Expected December 2027

Bachelor of Science in Applied Data Science

- BlueDot Impact Technical AI Safety course (June 2026 – Present): interpretability, safety, model behaviors & evals

Research Experience

Look Before You Steer: Geometry Predicts SAE Feature Steerability [paper] June 2026

Co-First Author, Algoverse AI Research *ICML 2026 Mech. Interp. Workshop*

Can you tell which sparse autoencoder (SAE) features will steer a model well, before trying them?

- Showed that a feature's position among the other features (how crowded its neighborhood is, how close its nearest neighbor is) predicts steering success before any steering is run (Spearman ρ up to -0.55, $p < 10^{-6}$; AUROC up to 0.82)
- Held across 4 models (Gemma-2 2B/9B, Llama-3.1-8B, Qwen3-8B) and 2 SAE sizes; mapped the layers where it weakens
- Built the evaluation: 100+ features × 100 prompts × 9 steering strengths, with coherence checks, refusal scoring on SALAD-Bench and shuffled-label tests to rule out chance. Batching cut model calls by 94% (69K to 4.3K)

Latent Knowledge Readout September 2026 – Present

Undergraduate Researcher, Penn State, advised by Dr. Lu Lin

What does a model's internal state reveal about whether it will fail a task, or act against what it knows?

- Trained probes on Qwen3-1.7B's internal state over 4,264 attempts at a text crafting game: they beat a transcript-reading classifier at spotting invalid next actions (AUC 0.97–0.99 vs. 0.83–0.84) but not at predicting task failure (within 0.03)
- Pre-registered tests on agent benchmarks for household tasks (Qwen2.5-7B) and tool use (Qwen3-8B). On household tasks, a failure probe trained on late steps fell to chance on early steps of the same episodes (AUC 0.49–0.51 vs. 0.62–0.63)
- One prediction was wrong: probes for single-step errors did not carry over between early and late steps either
- Now studying what happens inside a model that knows the right answer but gives a wrong one when a user pushes back

Interpretability for AI Financial Advisors (Mechanistic KYC) August 2026 – Present

Independent Researcher, The Pennsylvania State University

When an AI financial advisor gives in to a pushy client, does its view of the client's risk change, or only its answer?

- Built 6,000 synthetic client profiles with known risk tolerance (Grable & Lytton scale) as test cases
- Planning probe and patching experiments on Gemma-2-9B and Llama-3.1-8B to separate its own risk attitude from the client's

Auditing Internal-State Probes for Privacy Leaks August 2026 – Present

Independent Researcher

Before a model answers, can its internal state show whether it will reveal private information?

- Compared probes on Qwen2.5-7B's internal state against text-only classifiers across 270 privacy scenarios (ConfAIde)
- Wrote a labeling guide and hand-checked 30 responses the judge marked "refused": 28 still revealed the private information
- Before it writes anything, the model's internal state predicts whether it will hedge or change the subject (AUROC 0.73, 0.16 above the best text baseline), with the layer picked without test data and a control for the model knowing it is tested
- Limits: the signal did not get stronger in the 8 tokens before a leak (pre-registered), and did not beat text on Llama-3.1-8B

Grants

BlueDot Impact Rapid Grant, $1,800 September 2026

BlueDot Impact Rapid Grant, $2,500 July 2026

Other Experience

Research Assistant January 2025 – June 2025

Humanitarian Engineering and Social Entrepreneurship (HESE), Penn State

- Analyzed water-sensor data to adjust feeding schedules on a fish-farming project (Hexarech); fish deaths fell 30–50%
- Co-designed low-cost fish cages with local fisheries groups and hatcheries during fieldwork in Kisumu, Kenya